Suggestions for Further Reading

Deacon, T. W. (1997). *The symbolic species.* New York: W. W. Norton and Co. A brilliant, profound analysis of human language and how it relates to the evolution of the human brain.

Pinker, S. (1994). *The language instinct.* New York: William Morrow and Company. Clear, often entertaining discussion of the psychology of language.

Terms

productivity the ability to express new ideas (page 297)

transformational grammar a system for converting a deep structure into a surface structure (page 297)

language acquisition device a built-in mechanism for acquiring language (page 300)

poverty of the stimulus argument the belief that children do not receive enough information from what they hear for them to learn or infer grammar, so they must be born with that knowledge (page 300)

Broca's aphasia a condition characterized by inarticulate speech and by difficulties with both using and understanding grammatical devices—prepositions, conjunctions, word endings, complex sentence structures, and so forth (page 300)

Wernicke's aphasia a condition marked by difficulty recalling the names of objects and impaired comprehension of language (page 300)

bilingual learning to use two languages about equally well (page 303)

word-superiority effect identifying a letter with greater ease when it is part of a whole word than when it is presented by itself (page 306)

phoneme a unit of sound (page 307)

morpheme a unit of meaning (page 307)

fixations periods when the eyes are stationary (page 307)

saccade a quick jump in the focus of the eyes from one point to another (page 307)

Answers to Concept Checks

8. Start language learning when a child is young. Rely on imitation as much as possible, instead of providing direct reinforcements for correct responses. (page 299)
9. Children begin to string words into novel combinations as soon as they begin to speak two words at a time. We believe that they learn rules of grammar because they overgeneralize those rules, creating such words as *womans* and *goed*. (page 302)
10. Set up a Stroop task, asking the person to read off the color of the ink of all the words. Let the words be color names from many languages. The person should show some delay at naming the colors of ink for English words (*red, green,* and so forth) and for whatever other language he or she knows well. (page 303)
11. The word *thoughtfully* has seven phonemes: th-ough-t-f-u-ll-y. (A phoneme is a unit of sound, not necessarily a letter of the alphabet.) It has three morphemes: thought-ful-ly. (Each morpheme has a distinct meaning.) (page 307)
12. Two or three short words can fall within the "window" of about 11 letters that we can fixate on the fovea at one time. If the words are longer, it may be impossible to get them all onto the fovea at once. (page 309)
13. Probably, but not always. Suppose your eyes fixate on the fourth letter of *memorization*. You should be able to see the three letters to its left and the seven to its right—in other words, all except the final letter. Because there is only one English word of the form *memorizatio-*, you have enough information to recognize the word. (page 309)

Web Resources

Chimpanzee and Human Communication Institute

www.cwu.edu/~cwuchci/main.html

Visit Central Washington University's Chimpanzee and Human Communication Institute (CHCI), and learn more about Washoe, her "adopted son" Loulis, and three other chimps who have learned to use American Sign Language to communicate with humans. You might even want to apply for the 10-week Summer Apprenticeship Program.

The Gorilla Foundation

www.gorilla.org/GF/index.html

The Gorilla Language Project, or Project Koko, is the longest continuous interspecies communication project of its kind in the world. Koko, Michael, and Ndume are the western lowland gorillas involved in this study of language acquisition.

Intelligence and Its Measurement

9

The famous mathematician Alan Turing bicycled to and from work each day. Occasionally, the chain fell off his bicycle and he had to replace it. Eventually, Turing began to keep records and noticed that the chain fell off at mathematically regular intervals. In fact, it fell off after exactly a certain number of turns of the front wheel. Turing then calculated that this number was an even multiple of the number of spokes in the front wheel, the number of links in the chain, and the number of cogs in the pedal. From these data, he deduced that the chain came loose whenever a particular link in the chain came in contact with a particular bent spoke on the wheel. He identified that spoke, repaired it, and never again had trouble with the bicycle chain (Stewart, 1987).

Turing's solution to his problem qualifies as highly intelligent, according to what we usually mean by *intelligent*. But hold your applause. Your local bicycle mechanic could have solved the problem in just a few minutes, without using any mathematics at all.

So, you might ask, what's my point? Was Turing unintelligent? Not at all. He was highly intelligent. If you have a new, complicated, unfamiliar problem to solve, you should probably take it to someone like Turing, not to your favorite bicycle mechanic.

My point is that intelligence is a combination of general abilities and practiced skills. The term *intelligence* can refer to the highly practiced skills shown by a good bicycle mechanic, a Micronesian sailor, a hunter-gatherer of the Serengeti Plain, or any other person with extensive experience and special expertise in a particular area. *Intelligence* can also refer to the generalized problem-solving ability that Turing displayed—the kind of ability that one can apply in an unfamiliar situation. But even that sort of ability develops gradually, reflecting the contributions of many kinds of experience.

The term *intelligence* refers to both generalized problem solving and to highly practiced skills.

Intelligence and Intelligence Tests

What is intelligence?

What is the purpose of IQ tests?

What do the scores on IQ tests mean?

Intelligence testing has a long history of controversy, partly because of misconceptions about its purpose. Consider this analogy: You have just been put in charge of choosing the members of your country's next Olympic team. However, the Olympic rules have been changed: Each country can send only 30 men and 30 women, and each athlete must compete in every event. Furthermore, the competitive events will be new ones, not exactly like any of the familiar events, and the Olympic Committee will not publish the rules for any of the new events until all of the athletes have arrived at the Olympic site. Clearly, you cannot hold the usual kind of tryouts, but neither will you choose people at random. How will you proceed?

Your best bet would be to devise a test of "general athletic ability." You might measure the abilities of all the applicants to run, jump, change direction, maintain balance, throw and catch, kick, lift weights, respond rapidly to signals, and perform other athletic feats. Then, you would choose the applicants who had the best scores, confident that they will do reasonably well at whatever events the Olympic Committee selects.

No doubt, your test would be imperfect. But if you must choose 60 athletes, and if you want to maximize their chances of winning, you must certainly use some sort of test. So, you go ahead with your test of general athletic ability.

As time passes, other people begin to use your test, and it becomes well accepted and widely used. Does its acceptance imply that athletic ability is a single quantity, like speed or weight? Not at all. For certain purposes, you found it useful to act as if athletic ability were a single quantity, even though you know that great basketball players are not necessarily great swimmers or gymnasts.

What Is Intelligence?

Intelligence tests resemble our imaginary test of athletic ability. If you were in charge of choosing which applicants to admit to a school or college, you would want to select those who would profit most from the experience. Because students may be studying subjects that they have never studied before, it makes sense to measure their general ability to profit from education rather than any specific knowledge or specialized ability.

Intelligence tests were developed for the practical function of selecting students for admission or placement in schools. These tests were not based on any theory of intelligence, and those who administer them have often been content to define *intelligence* as the ability to do well in school. Given that definition, IQ tests do measure intelligence.

For theoretical purposes, however, that definition is hardly satisfactory. What would be a better definition? Here are some of the ways that psychologists have defined *intelligence* (Sternberg, 1997; Wolman, 1989):

- The mental abilities that enable one to adapt to, shape, or select one's environment.
- The ability to judge, comprehend, and reason.
- The ability to understand and deal with people, objects, and symbols.
- The ability to act purposefully, think rationally, and deal effectively with the environment.

Note that these definitions use such terms as *judge, comprehend, understand,* and *think rationally*—terms that are themselves only vaguely defined. Psychologists would like to better organize this list of intelligent abilities, so that it has some structure or organization. Just as all the objects in the world are composed of compounds of only 92 elements, most psychologists expect to find that all the various kinds of intelligence are compounds of a few basic abilities. They have proposed several models of how intelligence is organized.

Spearman's Psychometric Approach and the g Factor

The **psychometric approach** to intelligence, pioneered by Charles Spearman (1904), features ***the measurement (metric) of individual differences in behaviors and abilities.*** Spearman began by measuring how well a variety of people performed a variety of tasks, such as following complex directions, judging musical pitch, matching colors, and performing arithmetic calculations. He then found that performance on any single task correlated positively with performance on all the other tasks. He therefore deduced that all these tasks must have something in common. To perform well on any test of mental ability, Spearman argued, people need a certain *"general" ability,* which he called ***g.*** (The *g* is always italicized and always lowercase.) Later researchers have confirmed that scores on virtually all kinds of cognitive tests correlate fairly strongly with one another, including tests that do not require any particular knowledge or schooling. Consider, for example, the following task: Either  or  is flashed on a screen for a fraction of a second. Then,  is flashed on the screen in the same location, to serve as a "masking stimulus" (to make the task a bit more difficult). The observer's task is to say whether the left or right arm was longer on the first stimulus. The investigator varies the duration of the first stimulus to determine the briefest flash that still enables the observer to answer correctly. Performance on this perceptual task has a correlation of at least .4 to .5 with scores on a standard intelligence test (Deary & Stough, 1996). So, intelligence tests are measuring something that shows up in a wide variety of situations.

To account for the fact that performance on various tasks does not correlate perfectly, Spearman suggested that each task requires the use of a *"specific" ability,* **s,** in addition to the general ability, *g,* that all tasks require (Figure 9.1). Thus, intelligence consists of a general ability plus an unknown number of specific abilities, such as mechanical, musical, arithmetical, logical, and spatial ability. Later research has found that many of these specific abilities develop somewhat independently and may rely on partly separate genetic influences (Loehlin, Horn, & Willerman, 1994; Pedersen, Plomin, & McClearn, 1994). Spearman called his theory a "monarchic" theory of intelligence because it included a dominant ability, or monarch (*g*), which ruled over the lesser abilities.

Psychologists do not entirely agree on what *g* represents. That is, the ability to do well on one task does correlate with the ability to do well on another task, but that correlation could indicate either a single, unitary process—such as the ability to perceive and manipulate relationships—or else the correlation could represent the fact that many separate processes depend on the same growth factors (Petrill, Luo, Thompson, & Detterman, 1996). Consider the following analogy:

First, consider the high correlation among the tasks shown in Figure 9.2: People who excel at running a 100-meter race also generally do well at the high jump and the long jump. A particular athlete might be a little better at one of these events than the others, but we can hardly imagine an outstanding high jumper who could not manage a decent long jump. The reason for this high correlation is that all

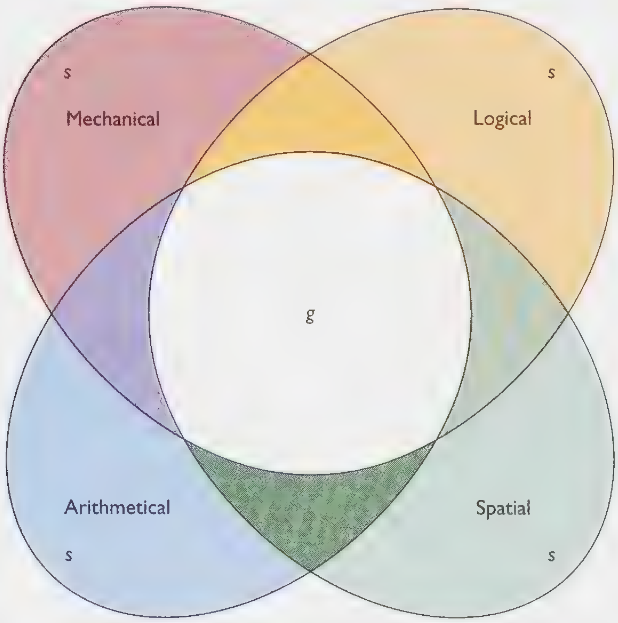


FIGURE 9.1 According to Spearman (1904), all intelligent abilities have an area of overlap, which he called *g* (for "general"). Each ability also depends partly on an *s* (for "specific") factor.

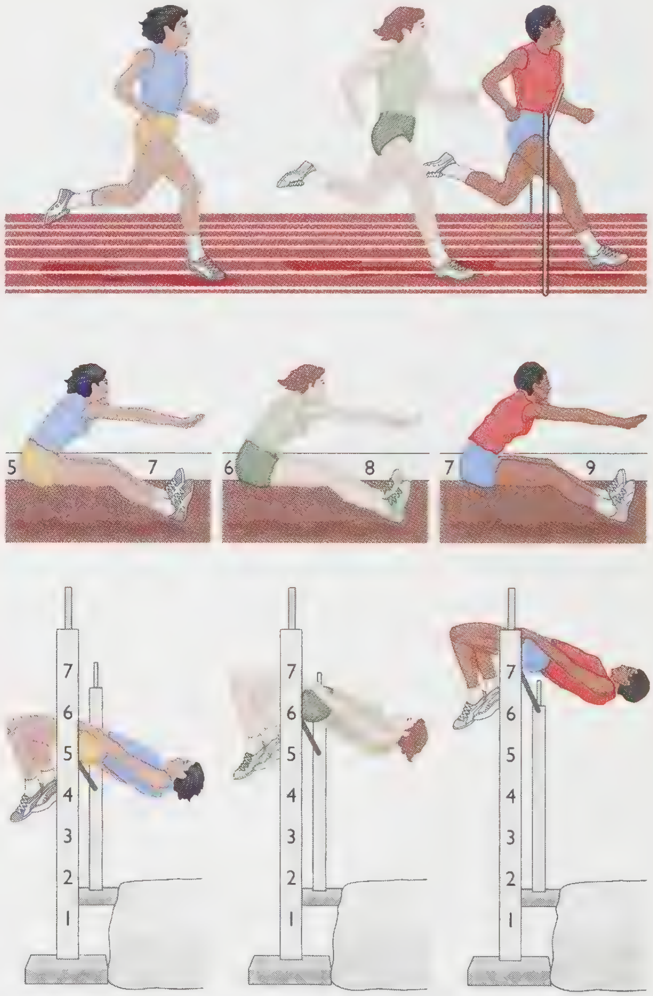


FIGURE 9.2 Measurements of sprinting, high jumping, and long jumping correlate with one another because they all depend on the same leg muscles. Similarly, the *g* factor that emerges in IQ testing could reflect a single ability that all tests tap.

FIGURE 9.3 Measurements of leg, arm, and finger length correlate with one another only because the genes, nutrition, and health factors that promote the growth of any one of these also promote the growth of the others. Similarly, the *g* factor that emerges in IQ testing could reflect that different intellectual abilities depend on the same nutritional, health, and educational influences.

three events depend on the same leg muscles. An injury that impaired performance on one event would impair performance on all three.

Second, consider the lengths of three body parts—the left leg, the right arm, and the left index finger—as illustrated in Figure 9.3: Within a normal population of people, these three measurements will correlate strongly with one another. That is, people with a long left leg also tend to have a long right arm and a long left index finger. Why? These measurements definitely do not measure the same thing. (Amputation of one of them would not affect the others at all.) Lengths of leg, arm, and finger correlate because the *causes* of growth are the same for all three—nutrition, health, age, and certain genes.

Now, which of these cases is most like intelligence? Do we find a positive correlation among the performances of various intellectual skills because they all measure a single underlying ability or because they all grow together, dependent on the same factors of health, nutrition, education, genetics, and so forth?

The evidence suggests that both of these hypotheses are partly correct. For example, certain kinds of brain damage impair performance on a wide variety of intellectual tasks, because the damage impairs a function such as attention that is necessary for almost any task. Thus, various intellectual abilities will correlate with one another partly because they depend on some of the same underlying processes. However, as we have seen in previous chapters, some people can lose a specific kind of memory or perceptual ability without much loss of other abilities. That is, different intellectual abilities do not share *all* of their underlying processes. To a considerable degree, various intellectual skills correlate with one another because health, nutrition, education, and the other factors that promote the development of any one skill also promote the development of others.

Fluid Intelligence and Crystallized Intelligence

Raymond Cattell accepted Spearman's psychometric approach but proposed one important modification. According to Cattell (1987), the *g* factor has two components: fluid intelligence and crystallized intelligence. Thus, intelligence can be compared to water: Fluid water can take any shape, whereas ice crystals are rigid. **Fluid intelligence** is *the power of reasoning and using information.* It includes the ability to perceive relationships, solve unfamiliar problems, and gain new types of knowledge. Fluid intelligence mostly relates to the ability to store large amounts of information in working memory and then to process that information quickly (Fry & Hale, 1996). (If you cannot process the information quickly enough, it will start to fade from your working memory.) **Crystallized intelligence** consists of *acquired skills and knowledge and the application of that knowledge to the specific content of a person's experience.* For example, fluid intelligence is the ability to learn new skills in a new job. Crystallized intelligence includes the skills already learned and practiced by someone who has had the job for years. Eventually those skills become almost automatic.

Fluid intelligence, according to Cattell and his colleagues, reaches its peak before age 20; beyond that age, it remains fairly constant, although it can decline sharply as a result of Alzheimer's disease or any other deteriorative condition or source of brain damage (Horn, 1968). Crystallized intelligence, on the other hand, continues to increase as long as a person remains healthy and active (Cattell, 1987; Horn & Donaldson, 1976). A 20-year-old may be more successful than a 65-year-old at solving a problem that is unfamiliar to both of them, but the 65-year-old will excel at solving problems in his or her area of specialization.

Although the distinction between crystallized and fluid intelligence is useful, it is not an absolute. Some tasks require mainly crystallized or mainly fluid intelligence, but any task taps at least a little of both.

CONCEPT CHECK

1. Was Alan Turing's solution to the slipping bicycle chain (at the start of this chapter) an example of fluid or crystallized intelligence? Would the solution provided by a bicycle mechanic be an example of fluid or crystallized intelligence? (Check your answers on page 326.)

Gardner's Theory of Multiple Intelligences

The traditional defense of theories proposing a single kind of intelligence has rested on the concept of *g*: Tests of all kinds of intellectual abilities—mathematical, verbal, and others—correlate with one another and therefore support the idea of a single underlying ability called *g*. However, critics reply, the statistical emergence of *g* merely indicates that mathematics, language, and the other tested skills happen to be related. If we expand the concept of *intelligence* to include other skills that society values, the concept of *g* may fade away or even disappear.

In particular, Howard Gardner (1985) has claimed that people have **multiple intelligences**—*numerous unrelated forms of intelligence*. Gardner distinguishes language abilities, musical abilities, logical and mathematical reasoning, spatial reasoning, body movement skills, self-control and self-understanding, and sensitivity to other people's social signals. He points out that people can be outstanding in one type of intelligence but not in others. For example, a *savant* (literally, "learned one") is a person whose performance in one area is outstanding but whose performance in other areas is just ordinary, or perhaps well below average. One pair of twins were intellectually impaired in most regards and could not even do simple arithmetic, but they could perform amazing feats of calendar calculation. You could call out any date, past or future (such as August 28, 1591), and they would quickly state the day of the week—in this case, Wednesday (Horwitz, Kestenbaum, Person, & Jarvik, 1965).

More generally, an athlete can excel at body movement skills but lack musical abilities; an outstanding musician can be insensitive to other people; a politician can be outstanding at understanding other people's needs but quite inept at controlling his or her own body movements. Because of the differences among multiple kinds of intelligence, someone who seems very intelligent in certain regards may surprise us by doing something utterly foolish in a different setting.

Gardner certainly makes an important point: People do have a variety of socially valued abilities, and almost no one is strong in all abilities or weak in all abilities. Gardner encourages us to seek better ways of teaching and encouraging a full range of abilities, including self-control, social responsiveness, and other intelligent skills that educators have not traditionally emphasized.

Do we want to use one word, *intelligence*, to refer to every valued skill from writing a novel to dribbling a basketball? If we do, then Gardner is correct that people have many unrelated kinds of intelligence. However, in this case, we shall need a new term (perhaps *cognitive ability*) to refer more narrowly to mathematical, logical, and verbal skills.

Sternberg's Triarchic Theory of Intelligence

Spearman concluded that intelligence depends on one overall ability; Cattell suggested two kinds of ability, and Gardner suggested many. Still, the concept of ability tells us nothing about *how* a person processes information or engages in "intelligent" behavior (Das, 1992).

Robert Sternberg (1985) has offered one of the most influential attempts to specify in detail the processes of intelligent behavior. Sternberg's description is called a **triarchic theory** (in contrast to Spearman's monarchic theory), because Sternberg deals with *three aspects of intelligence: (1) the cognitive processes that occur within the individual, (2) the situations that require intelligence, and (3) how intelligence relates to the external world.*

According to Sternberg, the first part of the triarchy—the cognitive processes within the individual—includes three components: learning the necessary information,

According to Howard Gardner, we have many intelligences, including mathematical ability, artistic skill, muscle skills, and musical abilities.

In Garrison Keillor's fictional *Lake Wobegon*, "all the children are above average." Although that description sounds impossible, in one sense it is true: Everyone is above average at something. A given individual may excel at mathematics, dancing, piano playing, juggling, poetry, cooking, or whatever. A single measurement of "intelligence" necessarily overlooks people's specialized skills.

planning an approach to a problem, and combining the knowledge with the plan to actually solve a problem.

Standard IQ tests do not try to measure the learning or planning components; these concentrate on the third component—how well someone actually solves a problem. For example, a test might ask you to complete an analogy, such as

> Washington is to 1 as Lincoln is to:
> (a) 5, (b) 10, (c) 20, (d) 50.

To solve this problem, you must have already learned a fair amount of relevant information, and you must develop a strategy for applying it to this question. Ultimately, if you are successful, you decide that the question must be referring to George Washington and the $1 bill and that the answer is a, because Abraham Lincoln's picture is on the $5 bill. This question is reasonable, as far as it goes, although we need to recognize that it measures only the final outcome, not the steps that led up to it. An improved measure of intelligence might separate the learning and planning components from the performance component.

The second part of Sternberg's triarchy is the identification of situations that require intelligence. It is important to distinguish novel situations from repeated situations, because they require different responses. In a novel situation, we must examine a problem in various ways until we find a successful approach. (Recall, for example, the insight problems of Chapter 8.) In a repeated situation, we profit by developing automatic habits, so that we can quickly make a successful response.

The third part of the triarchy is the relationship between intelligence and the outside world. An intelligent person either adapts to the environment or tries to improve the environment, or if all else fails, escapes to a better environment. Sternberg (1985) and others (Weinberg, 1989) have tried to develop special tests of people's practical intelligence—for example, the ability of a young person to identify the steps most likely to lead to career advancement. (See Figure 9.4.) Sternberg explicity recognizes that, because intelligent behavior must be practical, the term *intelligence* is meaningful only in a sociocultural context. For example, we cannot meaningfully compare the intelligence of a European city-dweller with that of someone living in the Amazon rain forests or the Serengeti Plain. What is important and practical for a member of one group may not be for the others.

Table 9.1 summarizes four theories of intelligence.

CONCEPT CHECK

2. In Sternberg's theory, novel situations call for creative problem solving, whereas repeated situations call for automatic habits. Using Cattell's terminology, which kind of situation calls for fluid intelligence and which calls for crystallized intelligence? (Check your answer on page 326.)

Theories of Intelligence and Tests of Intelligence

The standard IQ tests, which we shall consider momentarily, were devised decades ago, before most of the discoveries about memory and cognition that we discussed in the last two chapters. Today, there are several theories

FIGURE 9.4 In these two photos, which person is the supervisor and which is the worker? Robert Sternberg has used these photos to evaluate people's "practical intelligence"—their ability to understand nonverbal cues.

TABLE 9.1 Four Theories of Intelligence

THEORY	PRINCIPAL THEORIST	KEY IDEAS AND TERMS	EXAMPLES
Psychometric approach	Charles Spearman	*g* factor: general abstract reasoning ability common to various tasks	Perceiving and manipulating relationships
		s factor: specific ability required for a given task	Mechanical, verbal, spatial abilities
Fluid and crystallized intelligence	Raymond Cattell	Fluid intelligence: reasoning and using information; peaks in young adulthood	Finding a solution to an unfamiliar problem.
		Crystallized intelligence: acquired skills and knowledge; continues growing throughout life	Knowing how to play the piano, build a cabinet, write a novel, calculate a sales tax
Multiple intelligences	Howard Gardner	Intelligence includes all the abilities that one's society values	Music, social attentiveness, dancing, language skills, mathematics, etc.
Triarchic theory	Robert Sternberg	Cognitive mechanisms	Gaining knowledge, planning a strategy, actually solving a problem
		Situations that require intelligence	Novel situations; repeated situations
		Relationship to the environment	Adapting to one's environment, improving it, or escaping from it

about intelligence and a variety of intelligence tests, but these theories have little relationship to the tests.

Can we measure something—in this case, intelligence—without fundamentally understanding what it is? Possibly so; physicists measured gravity and magnetism long before they understood them theoretically. Maybe psychologists can do the same with intelligence.

But then again, maybe not. Physicists of the past measured not only gravity and magnetism, but also "phlogiston," a substance that, they later discovered, does not exist. Measurements of a poorly understood phenomenon are risky. Many psychologists are dissatisfied with the currently available intelligence tests, and some are working toward eventually producing a fundamentally better test. The available tests have both strengths and weaknesses, however. Let's examine some of these tests.

IQ Tests

You have no doubt spent a significant portion of your life taking tests. Some need for testing is clear, even if our society does overdo it. More students want to attend medical school than our medical schools could possibly accommodate; more people would like to be airplane pilots than the airlines can hire; more people would like to play on their college basketball teams than those teams can include. We must somehow determine which applicants are most likely to perform the task well.

Schools and colleges base their admissions decisions partly on students' grades and teachers' recommendations, but they know that the value of such information is limited. (Some schools are better than others; some teachers grade more strictly than others.) To compare students from different schools, admissions officials also look at students' scores on standardized tests.

Intelligence quotient (IQ) tests attempt to *measure an individual's probable performance in school and similar settings.* (The term *quotient* dates from the time when IQ was determined by dividing *mental age* by *chronological age.* That method is now obsolete, but the term remains.) The first IQ tests were devised for a practical purpose by two French psychologists, Alfred Binet and Theophile Simon (1905). The French Ministry of Public Instruction wanted a fair way to identify children who had such serious intellectual deficiencies that they could not succeed in the public school system and who therefore should not be placed in the same classes with other students. The task of identifying such children had formerly been left to medical doctors; however, doctors had different standards, and the school system was looking for a fair, impartial test. Binet and Simon produced a test to measure the skills that children need for success in school, such as understanding and using language, computational skills, memory, and the ability to follow instructions.

Their test, and others like it, do make reasonably accurate predictions. But suppose that a test correctly predicts that one student will perform better than another in school. Can we then say that the first student did better in school *because of* a higher IQ score?

No. Consider this analogy: Suppose we ask why a certain basketball player misses so many of his shots. Someone answers, "Because he has a low shooting average." Clearly, that explains nothing. (The reason for his low shooting average is that he misses so many shots.) Similarly, saying that a student does poorly in school because of a low IQ score isn't much of an explanation; after all, the IQ test was designed to measure the very skills that schoolwork requires. An IQ score measures performance; it does not explain it.

The Stanford-Binet Test

The test that Binet and Simon designed was later modified for English speakers by Lewis Terman and other Stanford psychologists and published as the **Stanford-Binet IQ test.** This test is administered to individual students by someone who has been carefully trained how to present the items and how to score the answers. It contains items that range in difficulty, designated by age (see Table 9.2). An item designated as "age 8," for example, will be answered correctly by 60–90% of all 8-year-olds. (A higher percentage of older children will answer it correctly, as will a lower percentage of younger children.) Those who take this test are asked only those items that are pegged at about their level of functioning. For example, the psychologist testing an 8-year-old might start with the items designated for 7-year-olds. Unless that child missed many of the items for 7-year-olds, the psychologist would simply give credit for all the 6-year-old items without bothering to test them. Presuming that the child answered nearly all of the 7-year-old items correctly, the psychologist would proceed to the items for 8-year-olds, 9-year-olds, and so forth, until the child began to miss item after item. At that point, the psychologist would end the test without proceeding to the still more difficult items.

TABLE 9.2 Examples of the Types of Items on the Stanford-Binet Test

AGE	SAMPLE TEST ITEM
2	Test administrator points at pictures of everyday objects and asks, "What is this?" "Here are some pegs of different sizes and shapes. See whether you can put each one into the correct hole."
4	"Why do people live in houses?" "Birds fly in the air; fish swim in the ______."
6	"Here is a picture of a horse. Do you see what part of the horse is missing?" "Here are some candies. Can you count how many there are?"
8	"What should you do if you find a lost puppy?" "Stephanie can't write today because she twisted her ankle. What is wrong with that?"
10	"Why should people be quiet in a library?" "Repeat after me: 4 8 3 7 1 4."
12	"What does *regret* mean?" "Here is a picture. Can you tell me what is wrong with it?"
14	"What is the similarity between high and low?" "Watch me fold this paper and cut it. Now, when I unfold it, how many holes will there be?"
Adult	"Make up a sentence using the words *celebrate, reverse,* and *appointment.*" "What do people mean when they say, 'People who live in glass houses should not throw stones' "?

SOURCE: Modified from Nietzel and Bernstein, 1987.

Ordinarily, the entire test lasts anywhere from 1 hour to 1½ hours. However, unlike most other IQ tests, the current edition of the Stanford-Binet imposes no time limit; people are allowed to think about each item for as long as they wish (McCall, Yates, Hendricks, Turner, & McNabb, 1989).

Stanford-Binet IQ scores are computed from tables set up to ensure that a given IQ score will mean the same at different ages. The mean IQ at each age is 100. A 6-year-old, with an IQ score of, say, 116, has performed better on the test than 84% of other 6-year-olds; similarly, an adult with an IQ score of 116 has performed better than 84% of other adults. The Stanford-Binet also provides subscores reflecting crystallized intelligence, abstract visual reasoning, and short-term memory (Daniel, 1997; McCall et al., 1989).

In Table 9.2, note that the Stanford-Binet test includes questions designated for children who are only 2 years old. However, scores below age 4 or 5 fluctuate markedly (Honzik, 1974; Morrow & Morrow, 1974), because younger children are sometimes attentive to a task and sometimes not. The test scores of very young children are useful for certain research purposes, but they should not be weighed heavily when making placement decisions about individual children.

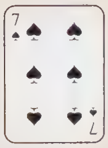
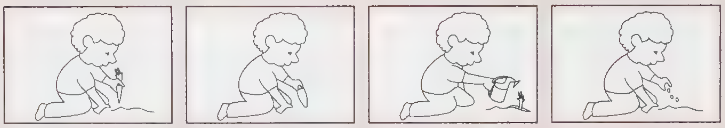
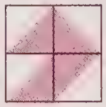
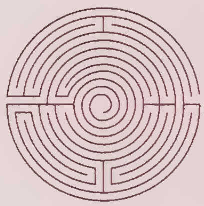
The Wechsler Tests

Two IQ tests devised by David Wechsler, known as the **Wechsler Adult Intelligence Scale–Third Edition (WAIS–III)** and the **Wechsler Intelligence Scale for Children–Third Edition (WISC–III),** produce the same average, 100, and almost the same distribution of scores as the Stanford-Binet. The WISC is given to children up to age 16; beyond that age, everyone takes the WAIS. As with the Stanford-Binet, the Wechsler tests are administered to one individual at a time. A Wechsler test provides an overall score and scores in two major categories (verbal and performance); each of these is further divided into component abilities. (Table 9.3 shows examples of test items, and Figure 9.5 shows one individual's test profile.) Thus, a Wechsler test provides a profile of the individual's strengths and weaknesses. People who learned English as a second language, for example, ordinarily receive higher scores on the performance section than on the verbal section.

Each of the 12 parts of the WISC–III or the WAIS–III begins with the simplest questions and progresses to increasingly difficult items. Six of the parts constitute the Performance Scale; these call for nonverbal answers (Figure 9.6), although the test-taker must know enough English to understand the instructions. The other parts, constituting the Verbal Scale, require spoken or written answers.

The inclusion of questions that ask for factual information (such as "From what animal do we get milk?") has caused much controversy. Critics complain that such items measure knowledge, not ability. Defenders reply as follows:

TABLE 9.3 Examples of the Types of Items on the Wechsler Intelligence Scale for Children (WISC–III)

TEST	EXAMPLE
Verbal Scale	
Information	From what animal do we get milk? (Either "cow" or "goat" is an acceptable answer.)
Similarities	How are a plum and a peach similar? (Correct answer: "They are both fruits." Half credit is given for "Both are food." or "Both are round.")
Arithmetic	Count these blocks: 
Vocabulary	Define the word *letter.*
Comprehension	What should you do if you see a train approaching a broken track? (A correct answer is "Stand safely out of the way and wave something to warn the train." Half credit is given for "Tell someone at the railroad station." No credit is given for "I would try to fix the track."
Digit Span	Repeat these numbers after I say them: 3 6 2.
Performance Scale	
Picture completion	What is missing from this picture? 
Coding	Here is a page full of shapes. Put a slash (/) through all the circles and an × through all the squares. △ ○ □ ▱ ○ △ ▱ □ ○
Picture arrangement	Here are some cards with a gardener on them. Can you put them in order? 
Block design	See how I have arranged these four blocks? Here are four more blocks. Can you arrange your blocks like mine? 
Object assembly	Can you put these five puzzle pieces together to make a dog? 
Mazes	Here is a maze. Start with your pencil here and trace a path to the other end of the maze without crossing any lines. 

SOURCE: Based on Wechsler, 1991. Items shown are similar but not identical to those actually on the WISC–III.

- "Intelligent" people tend to learn more facts than others do, even if they have had no more exposure to the information.
- As researchers in artificial intelligence have discovered, most of what we call intelligence requires a vast store of factual knowledge (Schank & Birnbaum, 1994). That is, factual knowledge is not *sufficient* to demonstrate intelligence, but it is *necessary.*
- The purpose of an IQ test is to predict performance in school, and in that respect it works. Students who already know a great many facts tend to do well in school.

We shall consider further criticism of intelligence tests in the second module of this chapter.

Raven's Progressive Matrices

The Stanford-Binet and Wechsler tests, though useful for many purposes, have certain limitations. First, they call for specific information that may be much more familiar to some people than to others. Second, because they require use and comprehension of the English language, they are

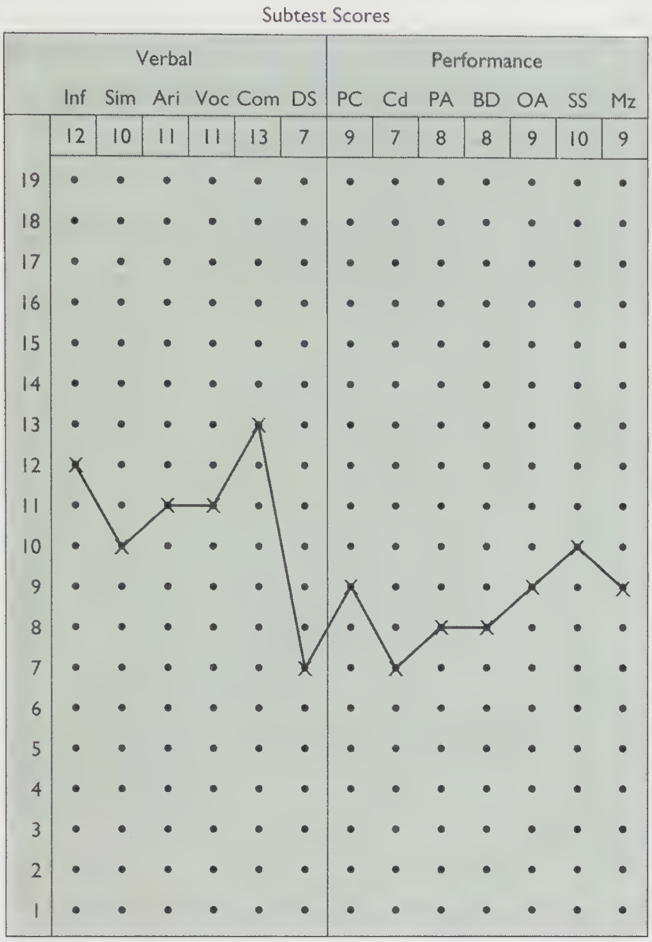


FIGURE 9.5 A score profile for one child on the WISC–III IQ test. Each subtest score represents this child's performance on one type of task compared with other children of the same age. Besides providing an overall IQ score, a profile such as this highlights an individual's strengths and weaknesses. Note that this child performed better on verbal tasks than on performance tasks. (Data courtesy of Patricia Collins.)

FIGURE 9.6 Much of the WAIS–III involves nonverbal tests where a person is asked to perform certain tasks. Here, to evaluate visual-spatial organization, a woman arranges colored blocks according to a specified pattern while a psychologist times her.

unfair to people who do not speak English well, including immigrants and hearing-impaired people. "Why not simply translate the tests into other languages?" you might ask. Psychologists sometimes do, but it is difficult to equate the translation with the original. For example, one part of the Stanford-Binet presents certain words and asks for words that rhyme with them. Generating rhymes is moderately easy in English, extremely easy in Italian, but, I am told, virtually impossible in Zulu (Smith, 1974).

To overcome such problems, psychologists have tried to devise a "culture-fair" or culture-reduced test that would make minimal use of language and would not ask for any specific facts. *One example of a culture-reduced test* is the **Progressive Matrices** test devised by John C. Raven. Figure 9.7 presents matrices of the type that this test uses.

TRY IT YOURSELF

These matrices, which progress gradually from easy items to difficult items, attempt to measure abstract reasoning; to answer the questions, a person must generate hypotheses, test them, and infer rules (Carpenter, Just, & Shell, 1990).

The Progressive Matrices test does not call for any verbal responses or specific information, and the instructions are easy to explain to almost anyone. It therefore gives a non-English-speaking immigrant or a deaf person a much better opportunity than most other IQ tests do (Powers, Barkan, & Jones, 1986; Vernon, 1967). The main disadvantage of this test is that it provides only a single overall score, instead of also identifying an individual's strengths and weaknesses, as other IQ tests do (Sternberg, 1991).

How culture-fair, then, is the Progressive Matrices test? It is fairer than the Wechsler or Stanford-Binet tests, but it does assume familiarity with pencil-and-paper, multiple-choice tests, and several other Western customs. For example, children from some cultures have been taught that it is disrespectful to express their own opinions without first checking with their parents, or not to speak to unfamiliar adults, such as the person administering the test (Greenfield, 1997). In Western culture, if a child is asked to sort items into categories, the response considered "most intelligent," for example, is to put all the metal tools in one category, all the foods in another, and so forth. But among the Kpelle people of Africa, the "intelligent" response would be to link objects that might be used together—such as placing a knife with a potato, because one would use the knife to cut the potato (Greenfield, 1997). In short, any test is based on the assumptions of a particular culture.

The Scholastic Assessment Test

The test once known as the Scholastic Aptitude Test, *designed to measure a student's likelihood of doing well in college*, is now titled the **Scholastic Assessment Test (SAT).** Because the term *assessment* means *test*, the title really means *school-based test test.* Many people call it the "SAT test," adding yet another layer of redundancy.

The Scholastic Assessment Test serves the same function as an IQ test: It predicts college performance. (Figure 9.8 shows the relationship between SAT scores and grade-point average during the freshman year in college at one university.) The basic SAT is administered to large groups of students at a time; it consists of multiple-choice items divided into two sets, verbal and quantitative. The SAT also offers subject tests in individual fields, such as history. Each test is scored on a scale from 200 to 800. The mean was originally set at 500. Over the years, the mean gradually drifted downward, partly because a wider range of students were taking the test, instead of just the best. In 1995, the scoring system was readjusted to return the mean to 500. (Thus, scores reported before the readjustment do not mean the same thing as scores reported after the readjustment.)

Figure 9.9 presents examples of the types of items found on the SAT. If you are a college student in the United States, you are probably familiar with either the SAT or the similar American College Test (ACT). Recent changes to the SAT have included the addition of a writing test and the option of taking the test on a computer.

The SAT was designed to help colleges to select among applicants for admission. Although the best predictor of college success is success in high school, high-school grades by themselves are not entirely satisfactory. Grading standards differ among various high schools, and some students take more challenging courses than others do. The SAT offers a method for comparing students from different high-school backgrounds. As a predictor of college success, the SAT by itself is less satisfactory than high-school grades alone. However, a combination of high-school grades and SAT scores significantly improves the prediction of college success (Weitzman, 1982).

Because many students worry about doing well on the SAT, a coaching industry has developed. Students can pay to attend these sessions after school or on weekends. If you attended such sessions, did you get your money's worth? If you did not attend them, did you miss a chance for a much higher SAT score? Several studies have compared the SAT scores of students who attended coaching sessions to the scores of students of similar ability who did not. The results have been consistent: On the average, participation in SAT-coaching sessions raises a student's score by about 10 to 20 points (on a scale from 200 to 800). Students who attend longer, more intensive coaching sessions score only slightly higher than those who attend briefer sessions. To improve scores by more than

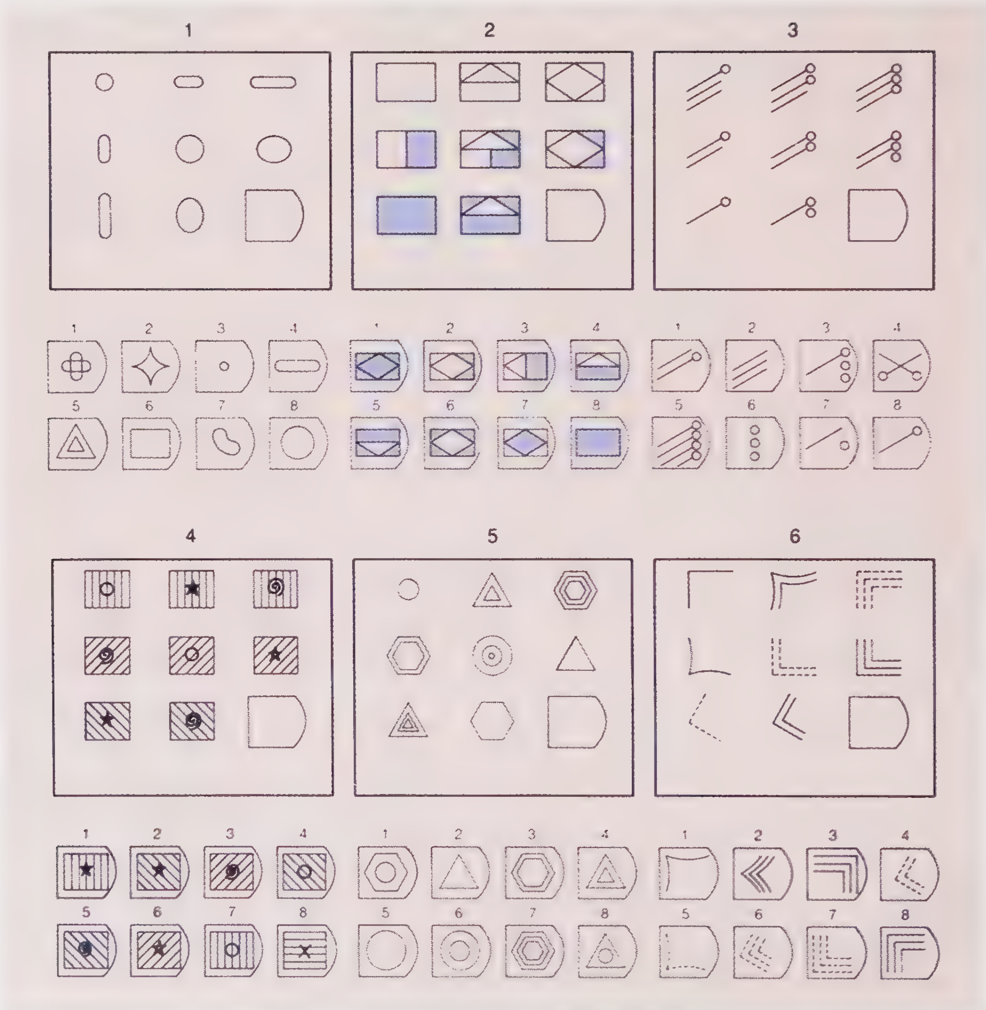


FIGURE 9.7 Items similar to those in Raven's Progressive Matrices test. The instructions are: "Each pattern has a piece missing. From the eight choices provided, select the one that completes the pattern, both going across and going down." (You can check your answers against answer A, page 326.)

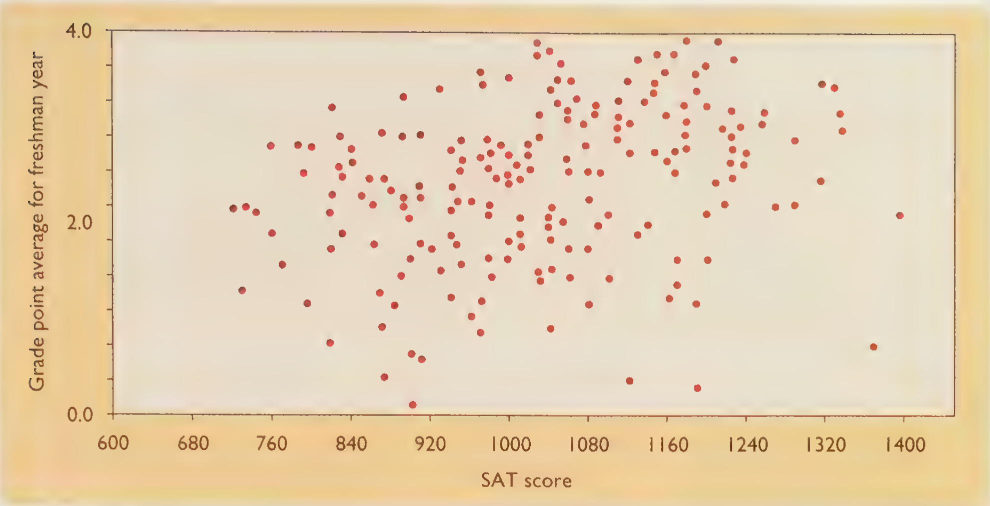


FIGURE 9.8 In this scatter plot, each point represents one freshman student at a particular university. That student's SAT score is given on the *x*-axis; the grade-point average, on the *y*-axis. Note that SAT scores predict college grades moderately well, with a correlation of .3 for this sample. Also note the exceptions—students with high SAT scores but poor grades and students with low SAT scores but good grades.

30 points, though, a student must spend almost as much time in the coaching sessions as attending school (Kulik, Bangert-Drowns, & Kulik, 1984; Messick & Jungeblut, 1981).

THE MESSAGE
Measuring Something We Don't Fully Understand

The standard IQ tests and related tests have survived for decades, despite persistent criticism. These tests have their pluses as well as their minuses. An IQ score is useful for certain practical purposes, such as selecting students for special programs. It also relates roughly to some biological variables. For example, studies using modern methods such as MRI scans have found a moderate, positive correlation between students' IQ scores and their brain volumes (Willerman, Schultz, Rutledge, & Bigler, 1991).

In a way, the overall intelligence of an individual is like the gross national product of a country; both figures provide a summary that may be useful for certain purposes. However, if we want to understand the processes in any detail, we need to go beyond those summaries and explore the detailed components that underlie them.

Psychologists hope to develop new tests that tap a wider range of abilities, including interpersonal (social) intelligence, creativity, practical intelligence, and learning style (Daniel, 1997). However, producing a significantly improved IQ test will not be as easy as it may sound. The current tests, even with their shortcomings, are the products of decades of research and effort; devising a valid measure of more complex and varied abilities will require at least as much research.

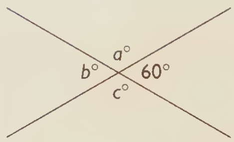
In the next module of this chapter, we shall consider how psychologists evaluate tests, including the currently available IQ tests and any new tests that might someday be proposed to take their place. Psychologists have devised some clear criteria for evaluating tests and deciding which tests are better than others.

At this point, let's simply stress that the value of an IQ test, like that of any tool, depends on how it is used. Just as a hammer can be used to build a door or to break one down, a test score can be used to open the doors of opportunity or to close them. A test score, if cautiously interpreted, can aid schools in making placement decisions. If the score is treated as an infallible guide, however, it can be seriously misleading.

SUMMARY

✱ *Defining intelligence.* The designers of the standard IQ tests defined intelligence simply as the ability to do well in school. Psychologists with a more theoretical interest have defined intelligence by listing the abilities that it includes. (page 315)

a



5. For the two intersecting lines above, which of the following must be true?

I. $a > c$
II. $a = 2b$
III. $a + 60 = b + c$

(A) I only
(B) II only
(C) I and II only
(D) II and III only
(E) I, II, and III

b

25. TYRANT : GOVERN ::
(A) inspector : examine
(B) inquisitor : question
(C) cynic : believe
(D) fugitive : escape
(E) volunteer : work

FIGURE 9.9 These two sample items from the Scholastic Assessment Test (SAT) reflect its two parts, which measure mathematical and verbal skills. (From the College Entrance Examination Board and the Educational Testing Service.)

* *g factor*. Various "intelligent" abilities apparently share a common element, known as the *g* factor, which is closely related to abstract reasoning. The *g* factor may arise either because various tests tap the same ability or because the health and educational factors that promote the growth of one intellectual ability also promote the development of other intellectual abilities. (page 315)
* *Fluid and crystallized intelligence.* Psychologists distinguish between fluid intelligence (a basic reasoning ability that can be applied to any problem, including unfamiliar ones) and crystallized intelligence (acquired abilities to solve familiar types of problems). (page 317)
* *Abilities that make up intelligence.* Psychologists have drawn up different lists of the abilities that make up intelligence. Some define intelligence fairly narrowly; others include such abilities as social attentiveness, musical abilities, and motor skills. According to the theory of multiple intelligences, people possess many independent types of intelligence. (page 318)
* *Triarchic theory.* According to Sternberg's triarchic theory of intelligence, intelligence consists of three aspects: the cognitive mechanisms within the individual, the situations that require intelligence, and the ways that intelligent behavior relates to the environment. (page 318)
* *IQ tests.* The Stanford-Binet and other IQ tests were devised to predict the level of performance in school. (page 320)
* *Wechsler IQ tests.* The Wechsler IQ tests measure separate abilities, grouped into a Verbal Scale and a Performance Scale of six parts each. (page 321)
* *Culture-reduced tests.* Culture-reduced tests such as Raven's Progressive Matrices can be used to test people who are unfamiliar with English. (page 322)
* *SAT.* The Scholastic Assessment Test is similar to IQ tests because it predicts performance in school, specifically in college. (page 323)

Suggestion for Further Reading

Ceci, S. J. (1990). *On intelligence . . . more or less.* Englewood Cliffs, NJ: Prentice-Hall. Perceptive critique of traditional assumptions about intelligence.

Terms

psychometric approach the measurement of individual differences in abilities and behaviors (page 315)

g Spearman's "general" factor that all IQ tests and all parts of an IQ test are believed to have in common (page 316)

s a "specific" factor that is more important for performance on some portions of an intelligence test than it is for others (page 316)

fluid intelligence the basic power of reasoning and using information, including the ability to perceive relationships, solve unfamiliar problems, and gain new types of knowledge (page 317)

crystallized intelligence acquired skills and knowledge and the application of that knowledge to the specific content of a person's experience (page 317)

multiple intelligences Gardner's theory that intelligence is composed of numerous unrelated forms of intelligent behavior (page 318)

triarchic theory Sternberg's theory that intelligence has three aspects: the cognitive processes that occur within the individual, the situations that require intelligence, and how intelligence relates to the external world (page 318)

intelligence quotient (IQ) a measure of an individual's probable performance in school and in similar settings (page 320)

Stanford-Binet IQ test a test of intelligence, the first important IQ test in the English language (page 321)

Wechsler Adult Intelligence Scale–Third Edition (WAIS–III) an IQ test originally devised by David Wechsler, commonly used with adults (page 321)

Wechsler Intelligence Scale for Children–Third Edition (WISC–III) an IQ test originally devised by David Wechsler, commonly used with children (page 321)

Progressive Matrices an IQ test that attempts to measure abstract reasoning without the use of language or recall of facts (page 323)

Scholastic Assessment Test (SAT) a test designed to measure a student's likelihood of performing well in college (page 323)

Answers to Concept Checks

1. Turing's solution reflected fluid intelligence, a generalized ability that could apply to any topic. The solution provided by a bicycle mechanic would reflect crystallized intelligence, ability developed in a particular area of experience. (page 318)
2. Using Cattell's terminology, a novel situation calls for fluid intelligence, whereas a repeated situation calls for crystallized intelligence. (page 319)

Answers to Other Questions in the Text

A. **1.** (8); **2.** (6); **3.** (3); **4.** (4); **5.** (6); **6.** (2) (page 324)

Web Resources

Barbarian's Online Test Page
www.iglobal.net/psman/intelligence.html
Take a deep breath before you try any of the dozen or so tests of intelligence on this page, because your brain is going to get a real workout!* You can also link to the EQ test on Barbarian's "Other Online Tests" page.

*Note: Any psychological test that is readily available to the public, while entertaining and possibly informative, should not be used to make important decisions about yourself or others. The most powerful, valid, and reliable psychological assessment devices are usually kept under the tight control of their creators or copyright holders.

MODULE 9.2

Evaluation of Intelligence Tests

What do scores on IQ tests mean?

What accounts for the variations in intelligence among different groups of people?

Edward Thorndike, a pioneer in the study of both animal and human learning, is often quoted as saying, "If something exists, it exists in some amount. If it exists in some amount, it can be measured." Douglas Detterman (1979) countered, "Anything which exists can be measured incorrectly."

Both of these quotes apply well to intelligence: If intelligence exists at all, it must be measurable, but it can also be measured incorrectly. One of the major tasks now facing researchers is to determine whether IQ tests measure what their designers claim they measure and whether they apply fairly to all groups. Because much is at stake here, the conclusions are often controversial.

The Standardization of IQ Tests

To specify what various scores mean, those who devise a test must *standardize* it. **Standardization** is *the process of establishing rules for administering a test and for interpreting the scores.* One of the main steps in standardization is to find the **norms,** which are *descriptions of the frequencies of occurrence of particular scores.*

Psychologists try to standardize a test on a large, representative population. For example, if a test is to be used with children throughout the United States and Canada, psychologists need to measure the norms with a large random or representative sample of U.S. and Canadian children, not just with children of one ethnic group or one geographic region.

The Distribution of IQ Scores

Binet, Wechsler, and the other pioneers who devised the first IQ tests chose items and arranged the scoring method to establish the mean score at 100, with a standard deviation of 15 for the Wechsler test, as Figure 9.10 shows, and 16 for the Stanford-Binet. (Recall from Chapter 2 that the standard deviation is a measure of the degree of variability of performance. If most scores are close to the mean, the standard deviation is small; if scores vary widely, the standard deviation is larger.) The scores of almost any population approximate a *normal distribution* or bell-shaped curve, as shown in Figure 9.10; the distribution is symmetrical, and most scores are close to the mean.

In any normal distribution, 68% of all people fall within one standard deviation above or below the mean; about 95% are within two standard deviations. Someone with a score of 115 on the Wechsler test exceeds the scores of people within one standard deviation from the mean, plus all of those more than one standard deviation below the mean—a total of 84% of all people, as shown in Figure 9.10. We say that such a person is "in the 84th percentile." Someone with an IQ score of 130 is in the 98th percentile, which means that his or her score is higher than the scores of 98% of others in the same age group.

Psychologists sometimes refer to people more than two standard deviations above the mean as "gifted." That designation is arbitrary. It makes little sense to say that a child with an IQ of 130 is gifted but a child with an IQ of 129 is not.

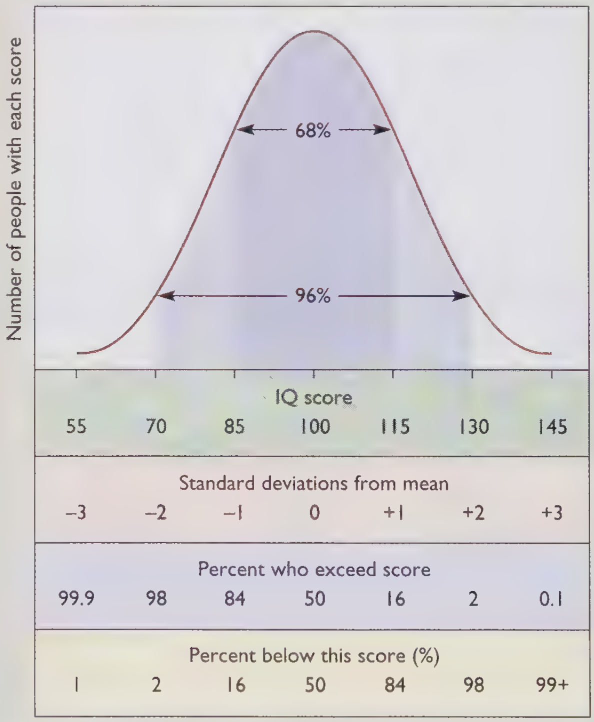


FIGURE 9.10 The scores on an IQ test form an approximately bell-shaped curve. The curve shown here represents scores on the Wechsler IQ test, with a standard deviation of 15 (15 points above and below the mean, which is 100). The results on the Stanford-Binet test are very similar, except that the standard deviation is 16, so the spread is slightly wider.

People with IQ scores at least two standard deviations below the mean are classified as "retarded." Many can be "mainstreamed" in regular classes; severely retarded children are taught in special classes.

Psychologists also classify people more than two standard deviations below the mean as "retarded." Many retarded children, especially those who are severely retarded, suffer from biological disorders, including chromosomal abnormalities and fetal alcohol syndrome (Zigler & Hodapp, 1991). In the United States, the *Individuals with Disabilities Act* requires public schools to provide "free and appropriate" education for all children, regardless of their limitations. Children with severe handicaps are placed in a special class, but those with milder handicaps are "mainstreamed" as much as possible, placed in the same classes with other children but given special consideration. For example, a child with a mild hearing impairment might sit close to the front of the room or a child with a motor handicap might be given extra time when taking tests. Unfortunately, we do not know as much as we wish we did about how to educate people with special needs (Detterman & Thompson, 1997).

Restandardization of IQ Tests

Over the years, the standardization of any IQ test eventually becomes obsolete. In 1920, a question that asked people to identify "Mars" was fairly difficult, because most people knew little about the planets. Today, in an era of space exploration, the same question would be very easy for most people. But this trend goes beyond just questions about factual material; most of the items on IQ tests seem to get easier from one decade to the next.

Consequently, the publishers of IQ tests periodically update them, replace easy questions with slightly harder ones, and make the scoring standards a bit stricter, to keep the mean score at 100. In other words, *decade by decade, generation by generation, people have been getting better and better at whatever it is that IQ tests are measuring.* If your generation were to take the same IQ test as your parents' generation, graded the same way, yours would score about 5–10 points higher, on the average. This tendency is known as the **Flynn effect,** after the psychologist who first called attention to it (Flynn, 1984, 1987). What accounts for the Flynn effect? Psychologists are not certain. Evolution is not a plausible explanation for such a rapid change. Improved health and nutrition are probably contributors, at least for improving scores at the lower end. Greater availability of education may be a factor, although the Flynn effect applies even to children 5–10 years old, and it seems doubtful that elementary schools have been getting steadily better from one decade to the next. One appealing hypothesis is that children have been receiving more and more benefits—yes, benefits—from television, movies, and computer programs (Neisser, 1997). Scores have been rising especially on highly visual tests, such as Raven's Progressive Matrices, which relate to the less verbal, more visual processing that television and other technologies promote. Regardless of the explanation, the Flynn effect demonstrates that changes in the environment can alter IQ scores.

Evaluation of Tests

At some point in your academic career, you probably complained that a test was unfair. Similarly, many people complain about intelligence tests. Much is at stake in this dispute; intelligence tests substantially influence the future of millions of people. Psychologists try to avoid simply arguing about whether or not a test appears to be reasonable; they examine specific kinds of evidence to determine whether the test achieves what it is intended to achieve. The main ways of evaluating any test are to check its reliability and validity.

Reliability

The **reliability** of a test is defined as *the repeatability of its scores* (Rogers, 1995). A reliable test measures something consistently. To determine the reliability of a test, psychologists calculate a correlation coefficient. They may test the same people twice, either with the same test or with equivalent versions of it, and compare the two sets of scores. Or they may compare the scores on the first and second halves of the test or the scores on the test's odd-numbered and even-numbered items. If all items measure approximately the same abilities, the scores on one set of items should be highly correlated with the scores on the other set of items. As with any other correlation coefficient, the reliability of a test can (theoretically) range from +1 to −1. In the real world, however, reliability is always positive. (A negative reliability would mean that most of the people who score high on one set of items score low on the other set. That pattern simply never happens.) Figure 9.11 illustrates **test-retest reliability,** *the correlation between scores on a first test and on a retest.*

FIGURE 9.11 On a test with high reliability, people who score high the first time will score high when they take the test again. On a test with low reliability, scores fluctuate randomly.

If a test's reliability is perfect (+1), the person who scores the highest on the first test will also score highest on the retest, the person who scores second highest on the first test will also score second highest on the retest, and so forth. If the reliability is 0, scores will vary randomly from one test to another. The reliability of the WISC–III has been measured at .96 (Wechsler, 1991), and the reliabilities of the Stanford-Binet, Progressive Matrices, and SAT are similar. These figures indicate that these IQ tests are measuring *something,* but they do not tell us what that something is or how useful it is.

CONCEPT CHECKS

3. I have just devised a new "intelligence test." I measure your intelligence by dividing the length of your head by its width and then multiplying by 100. Would that be a reliable test?
4. Most students find that their SAT scores increase the second time they take the test. Does that improvement indicate that the test is unreliable? (Check your answers on page 340.)

Validity

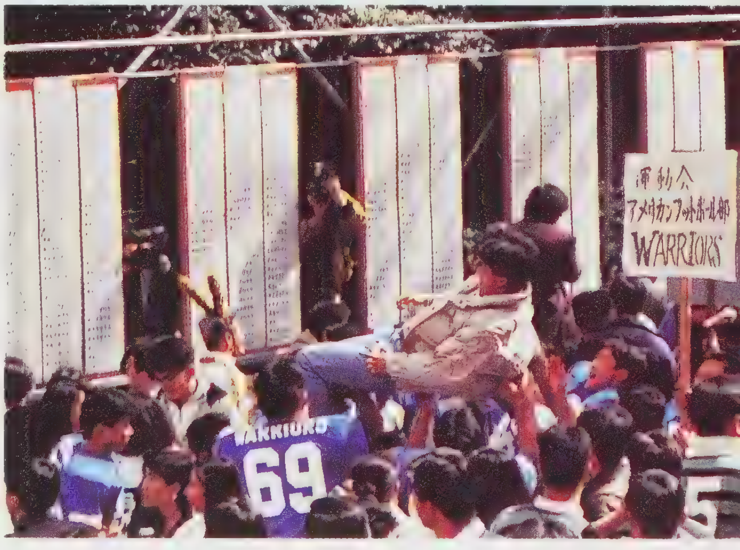
A test's **validity** is *a determination of how well it measures what it claims to measure.* One type of validity is **content validity.** We say that a test has high content validity if *its items accurately represent the information that the test is meant to measure.* For example, a test for a driver's license has content validity if it includes important laws and regulations that pertain to driving. A licensing examination for psychologists would have high content validity if it tested information that a practicing psychologist is expected to know.

A second type of validity is **construct validity.** A test has construct validity if *what it measures corresponds to a theoretical construct.* For example, intelligence is a theoretical construct that should have certain properties, such as increasing as a child grows older. For an IQ test to have construct validity, older children should, as a rule, answer more questions correctly than younger children do. It should also have the property of being independent of eyesight; a test would lack construct validity if blind people could not do well on it. In short, a test has good construct validity if the scores depend on factors we believe to be important and do not depend on factors we believe to be unimportant (Messick, 1995).

Predictive validity, a third type of validity, is *the ability of a test's scores to predict some real-world performance.* For example, an interest test that correctly predicts what courses a student will enjoy has predictive validity, as does an IQ test that correctly predicts how well a student will perform in school.

As with reliability, psychologists measure predictive validity by a correlation coefficient. To determine the predictive validity of an IQ test or the SAT, psychologists determine how well those scores predict students' grades. A validity of +1 would mean that the scores perfectly predicted performance; a validity of 0 would mean that the scores were worthless as predictors. A common problem in psychology is that the criterion that tests try to predict, such as grades in school, is itself an inaccurate measure of performance. The less accurately we measure the criterion, the harder it is for test scores to correlate highly with it.

The predictive validity of standard IQ tests and the SAT generally ranges from about .3 to .6, varying from one school to another (Anastasi, 1988; Siegler & Richards, 1982). As these figures suggest, success in school depends not only on academic skills, but also on motivation, persistence, and other personality attributes that are hard to measure—as well as what courses one takes, what teachers one has, and other factors that are impossible to predict.



In some countries, test scores determine a student's future almost irrevocably. Students who perform well are almost assured of future success; those who perform poorly will have very limited opportunities.

Validity of Test Scores for Selecting among Job Applicants Besides predicting school performance, IQ tests also have some validity (about .2 to .3) for predicting success in a variety of jobs (Wagner, 1997). To select people for advanced jobs, the best predictor is workers' current job performance or job knowledge. For entry-level jobs, however, tests of cognitive abilities help to identify workers who will learn a job quickly and well. According to Frank Schmidt and John Hunter (1981, p. 1128), "Professionally developed cognitive ability tests are valid predictors of performance on the job . . . for all jobs . . . in all settings." That is probably an overstatement. (It could hardly be an understatement!) Also, the controversy continues about whether job performance is more accurately predicted by more-or-less standard cognitive test scores or by measures of "practical knowledge" or common sense (Sternberg, Wagner, Williams, & Horvath, 1995). Personality factors such as persistence and willingness to listen and learn would probably be excellent predictors also, if we had better ways to measure them (Wagner, 1997). Still, for many jobs, using some type of cognitive test score to select employees increases the chances that those who are hired will succeed at their jobs.

In some cases, job applicants have challenged the right of an employer to deny them a job based on a test score or even based on their education. Their claim is that those criteria create unnecessary barriers for people who have little formal education but who can nevertheless perform the job well. U.S. courts have ruled that an employer can use test scores to select employees only if the employer can demonstrate that the scores are valid predictors of performance on that particular job. In fact, employers may use almost any criterion that they can demonstrate to be valid. If a police department wants to hire only tall police officers or an airline wants to hire only thin flight attendants, the courts ask a simple question (Hogan & Quigley, 1986): Can you demonstrate that people who meet this criterion do the job better than people who don't?

Special Problems in Measuring Validity Measuring the validity of a test can be tricky, especially at a school that relies heavily on the test scores for its admissions decisions. Consider data for the Graduate Record Examination (GRE), a test that is similar to the SAT. For predicting grades in the first year of graduate school, the verbal and quantitative parts of the GRE have the following predictive validities (Educational Testing Service, 1994):

	GRE verbal score	GRE quantitative score
Graduate students in Physics	.19	.13
Graduate students in English	.23	.29

Note that, for physics students, the verbal score is the better predictor of grades, whereas for English students, the quantitative score is a better predictor! How can we explain these surprising results? Simply. Almost all graduate students in physics have about the same (very high) score on the quantitative test, and almost all English graduate students have the same (very high) score on the verbal test. When almost all the students in a department have practically the same score, that score cannot predict which students will be more successful than others. A test can have a high predictive validity only for a population whose scores vary over a substantial range.

CONCEPT CHECKS

5. Can a test have high reliability and low validity? Can a test have low reliability and high validity?

6. If physics graduate departments tried admitting some students with low quantitative scores on the GRE and English departments tried admitting some students with low verbal scores, what would happen to the predictive validity of those tests?

7. Would you expect the SAT scores to show higher predictive validity at a university with extremely competitive admissions standards, such as MIT, or at a university that admits almost anyone who applies? (Check your answers on page 341.)

Utility

In addition to reliability and validity, a good test should have utility. **Utility** is defined as *usefulness for a practical purpose.* Not every test that is reliable and valid is also useful.

For example, one study found that first-year grades at the University of Pennsylvania correlated positively with both SAT aptitude test scores and SAT achievement test scores. That is, both aptitude scores and achievement scores were valid. However, the investigators found that they could predict first-year grades just as well from achievement scores alone as they could from a combination of aptitude and achievement scores (Baron & Norman, 1992). That is, if the university already had students' SAT achievement scores, the SAT aptitude test had little or no utility. Those results may not apply to other colleges, and they are of course irrelevant for colleges that do not require the achievement tests. The point is that any institution should determine whether a given test has utility for its purposes.

Table 9.4 summarizes the criteria for evaluating intelligence tests.

Interpreting Fluctuations in Scores

Suppose that your score on the first test in your Psychology course is 94% correct. On the second test (which was equally difficult), you achieve a score of only 88%. Does that score indicate that you studied harder for the first test than for the second test? Not necessarily. Whenever you take tests that are not perfectly reliable, your scores are likely to fluctuate. The lower the reliability, the greater the fluctuation.

TABLE 9.4 Evaluating Intelligence Tests

RELIABILITY	VALIDITY	UTILITY	BIAS
How consistent are the same person's scores?	How well does the test measure what it claims to measure? Content: Do the test items represent the pertinent information? Construct: Do the results match theoretical expectations? Prediction: Do the test scores predict real-world performance?	How useful is the test for a particular purpose?	Do test scores make equally accurate predictions from all groups?

When people lose sight of this fact, they sometimes offer complex explanations for random fluctuations in results. In one well-known study, Harold M. Skeels (1966) tested infants in an orphanage and identified those who had the lowest IQ scores. He then placed those infants in an institution that provided more personal attention. Several years later, most of those infants showed major increases in their IQ scores. Should we conclude, as many psychologists did, that the extra attention improved the children's IQ performances? Not necessarily (Longstreth, 1981). IQ tests for infants have low reliability—in other words, the scores tend to fluctuate widely. If someone selects a group of infants with low IQ scores and retests them later, the mean IQ score is almost certain to improve, simply because the scores had nowhere to go but up. Many of the infants were, no doubt, brighter than their original test scores indicated and received bad test scores because they were not feeling well, were not paying attention, or had other temporary problems.

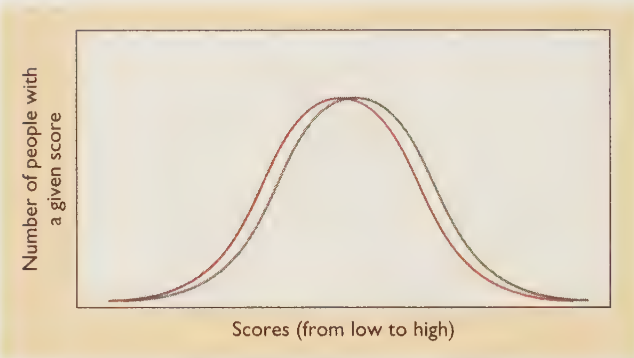
SOMETHING TO THINK ABOUT

What would be the proper control group for the study by Skeels? ✷

Group Differences in IQ Scores

Binet and the other pioneers in IQ testing discovered that girls tend to do better than boys on certain kinds of language tasks, especially verbal fluency. Boys, however, tend to do better than girls on visual-spatial rotations and certain kinds of mathematical reasoning. By loading the test with one type of item, they could have "demonstrated" that girls are smarter than boys or that boys are smarter than girls. Instead, they carefully balanced these two types of items to ensure that the mean score of both girls and boys would be 100.

Over the years, studies have continued to demonstrate the same kinds of male-female differences in many countries and cultures (Halpern, 1997). Note that these are differences between the *averages;* the distribution of male scores completely overlaps the distribution of female scores, and the differences within either group are large compared to the differences between the two groups:



Males tend to show greater individual variability. Thus, the data on a variety of intellectual measures show a higher percentage of males than females at the very top of the range and also at the very bottom of the range (Lubinski & Benbow, 1992). For example, one finds more males than females among the top achievers in many fields but also among those with mental retardation, attention deficit disorder, and reading and speaking impairments (Halpern, 1997). (*Why* this is true, we do not know.)

Ethnic groups in the United States also differ in their mean performances on IQ tests (Herrnstein & Murray, 1994). One problem when dealing with these data is that any group—such as "Black" or "African-American"—refers to a diverse population who vary among themselves both genetically and environmentally. Ordinarily, researchers classify people by simply asking them how they classify themselves; therefore, these categories are only partly satisfactory (at best) from either a biological or sociological standpoint. Nevertheless, these are the data, for whatever they are worth (Neisser et al., 1996; Suzuki & Valencia, 1997): The mean score of European-Americans as a whole is about 100, with the Jewish subpopulation averaging a few

Many immigrants to the United States settle in ethnic neighborhoods where they can use their original language. Most first-generation immigrants do not score highly on English-language intelligence tests; as a rule, their children and grandchildren get higher scores.

points above other European-Americans. The mean for African-Americans used to be 85 but has risen over the last several decades to about 88 or 90; African-Americans' scores on the SAT and other tests have risen also (Williams & Ceci, 1997). The mean score for Latinos is in the 90s, but many of those people are first-generation immigrants to the United States who are hampered by language problems; most Latinos score lower on the Verbal part of the Wechsler IQ test than on the Performance part, which relies less on language. The means for East Asians (those of Japanese, Chinese, or Korean descent) vary from one sample to another, sometimes about 100 and sometimes a few points higher. East Asians' school performance is even higher than one would predict from their test scores and is presumably the result of a culture that encourages diligent study.

Please bear in mind that these data reflect the means for entire populations. Each ethnic group includes many individuals with extremely high scores and many others with very low scores. These data do not justify prejudices or assumptions about any given individual.

What accounts for the observed differences among ethnic groups? In a famous but controversial book, *The Bell Curve*, Richard Herrnstein and Charles Murray (1994) argued that these IQ differences may be partly due to genetic differences. In response to this book and the ensuing controversy, a panel of distinguished investigators reviewed the evidence and issued a report (Neisser et al., 1996). The panel largely agreed with Herrnstein and Murray, with slightly different wording and emphases, about the following points:

- IQ tests, imperfect as they are, reliably measure something, regardless of whether we choose to call it intelligence.
- Whatever IQ tests measure correlates with school performance, the likelihood of getting certain jobs, and many other behavioral measures. It is not an all-important characteristic of a person, but it is not trivial either.
- Measured differences among groups are not illusions that we can attribute to faulty tests.
- IQ differences within the European-American population reflect a combination of environmental and genetic differences. Not much is known about the genes that affect intelligence. Important environmental factors include the fairly obvious ones such as education (Brody, 1997; Ceci & Williams, 1997; Perkins & Grotzer, 1997), but also such factors as early nutrition, exposure to lead and other toxins, and whether the mother drank alcohol during pregnancy.
- IQ differences among ethnic groups may also reflect a combination of environmental and genetic differences.

The first and second of those points say, in effect, that IQ tests are reliable and valid for certain purposes. As already mentioned earlier in this chapter, the data clearly support those two points. According to a consensus of 52 leading researchers in this area (Arvey et al., 1994), Herrnstein and Murray's third and fourth points also fit the data: The apparent group differences are not due to faulty testing, and differences among European-Americans are probably due to both genetic and environmental influences. On the fifth point, however, the evidence is too weak to justify any conclusion beyond saying that it *may* be true. The available evidence does not demonstrate a genetic basis for ethnic differences, but it also fails to eliminate that possibility. Quite simply, we do not have much strong evidence. With that preview in mind, let us proceed.

Should We Care?

On such a politically charged question, it is risky to say anything at all. Even discussing the differences between blacks and whites calls extra attention to such matters and runs the risk of supporting prejudices against minority groups. No doubt most of us look forward to the day when race and ethnicity are no longer a divisive issue and no one asks or much cares about IQ differences among groups; however, that day has not yet come. Many people do care about why blacks and whites differ, and many people form opinions about the issue, some of them very strong, frequently without knowing any of the evidence. If people are going to have opinions, they may as well know the evidence.

Are IQ Tests Biased?

One often-proposed explanation for group differences in IQ scores is that the tests are biased against certain groups. That is, ethnic groups do not actually differ in mean IQ; they merely appear to differ because of unfair or inappropriate tests. Presumably, according to this viewpoint, a better test would indicate no group differences.

The first point to emphasize here is that the existence of measurable group differences does not, by itself, demonstrate bias in the tests. If, on the average, one group gets

Students who return to school in their 30s or later usually get better grades than their test scores would predict. That is, the tests are "biased" against them in the technical sense of the term *bias*. Saying this does not mean the tests were designed to be unfair to them, just that the test scores underpredict their performance.

higher scores than the second group, it could be that the test is biased against the second group, but it is also possible that the groups really do differ in their performances. For example, on the average, 20-year-olds answer more questions correctly on almost any information test than do 6-year-olds. This fact does not indicate that information tests are biased against 6-year-olds; the difference in scores accurately reflects that 20-year-olds are better informed than 6-year-olds. If two groups differ in their performances, we need to collect evidence to find out whether the test is biased or not.

To determine whether or not a test is **biased** against a group, psychologists determine ***whether the test systematically underestimates group members' performance.*** To understand what it means for a test to be biased, consider two examples: First, the Stanford-Binet and Wechsler IQ tests are biased against non-English speaking immigrants to the United States. In the long run (after immigrants have learned the language), such people usually succeed beyond what their initial test scores would indicate. That is, the test scores underestimate the immigrants' eventual performance. Psychologists of an earlier era, either unaware of this bias or insensitive to it, drew some conclusions that now seem ludicrous about the "feeblemindedness" of immigrants (Gelb, 1986).

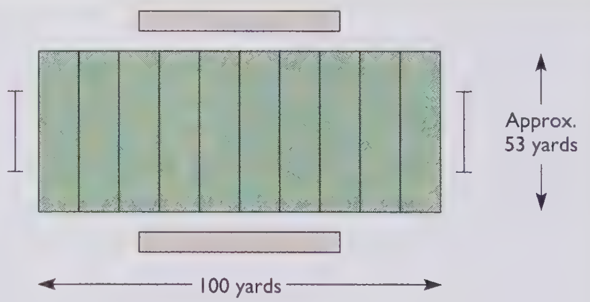
A second example: Women who enter or return to college in their 30s or later, generally receive better grades than their scores on the SAT, ACT, or any other test would predict. They also do better than their high-school grades would predict. When we say, therefore, that these tests are biased against such women, we are not suggesting that the test authors rigged the tests against them. We are merely saying that a given test score means something different for a 40-year-old woman than it does for a 19-year-old. Why do the women who return to school get better grades than their test scores predict? There are a couple of hypotheses: (1) Because they have been away from school for a while, their test-taking skills are rusty but will recover with a little practice. (2) To return to school at all at that point, they must have strong motivation.

To determine whether a test is biased—against middle-aged women, ethnic minorities, or whatever—psychologists must find out whether such individuals do better in school or college than the test scores predict. Psychologists try to identify bias both in individual test items and in the test as a whole.

Evaluating Possible Bias in Single Test Items

To determine whether a particular item on a test is biased, psychologists must go beyond an "armchair analysis" that says "this item seems unfair." They look for evidence that an item that is relatively easy for one group might be relatively difficult for another group (Schmitt & Dorans, 1990).

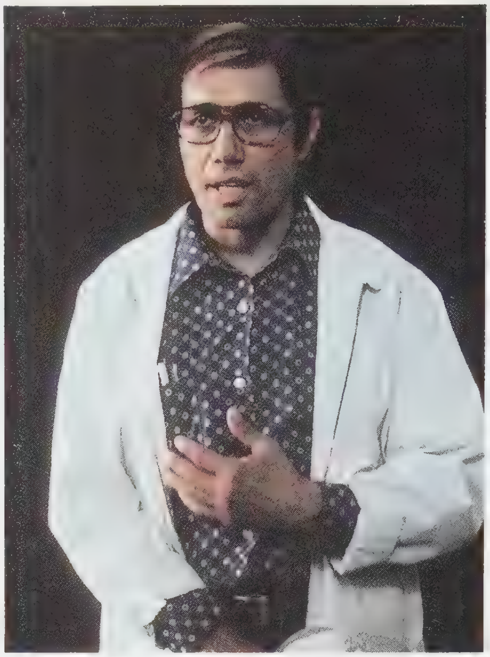
Figure 9.12, an item that once appeared on the SAT, shows a diagram of an American football field and asks for the ratio of the distance between the goal lines to the distance between the sidelines. For men, this was one of the easiest items on the test. The few men who missed it generally did poorly on the rest of the test as well. However, a higher percentage of women missed this item, including some women who did very well on the rest of the test. The reason was that some very bright women did not know which were the goal lines and which were the sidelines. This pattern of evidence indicated that the item was biased against women, and the publishers of the SAT therefore



The diagram above represents a football field. What is the ratio of the distance between the goal lines to the distance between the sidelines?

a. 1.89
b. 1.53
c. 0.53
d. 5.3
e. 53

FIGURE 9.12 This item was once included on the SAT, until psychologists determined that it was biased against women. Some women who did very well on the rest of the test did not know which were the goal lines and which were the sidelines.



"Students will rise to your level of expectation," says Jaime Escalante (left), the high-school teacher portrayed by Edward James Olmos (right), in *Stand and Deliver.* The movie chronicles Escalante's talent for inspiring average students to excel in calculus. School counselors warned that he asked too much of his students; parents said their kids didn't need calculus. And when his students first passed the advanced placement test, they were accused of cheating, a charge that seemed to reflect prejudice against the students, who were not European-American, middle-class, college-prep types.

removed the item from the test. As a rule, standardized tests such as the SAT are routinely purged of any items with demonstrable bias against a particular group.

Evaluating Possible Bias in a Test as a Whole

By definition, a biased test is one that systematically underestimates (underpredicts) the performance of a group. If an IQ test is indeed biased against African-Americans, for example, then African-Americans who score, say, 100 really have greater abilities than European-Americans with the same score and should outperform them in school.

The evidence, however, indicates that minority-group students with a given IQ score generally do about as well in school and at school-related tasks as do European-Americans with the same IQ score (Barrett & Depinet, 1991; Cole, 1981; Lambert, 1981; Svanum & Bringle, 1982). Likewise, minority-group students with a given SAT score generally do about as well in college as do European-Americans with the same scores (McCornack, 1983). The unpleasant fact is that, on the average, European-American students get better grades in school in the United States than African-Americans do. The IQ tests simply report that fact. In short, the tests show no evidence of ethnic-group bias, so continued claims of test bias could be described as "blaming the messenger" (the IQ tests) for the bad news.

Some critics have argued that African-American students perform more poorly than they might be expected to on various tests because they are either intimidated by a European-American tester or because they are confused by the language of the test. However, research studies have found no consistent increase in black children's test scores when the test is administered by an African-American examiner, using an African-American dialect (Sattler & Gwynne, 1982).

Saying that IQ tests show no demonstrable bias against minority groups does *not* mean, though, that the differences in scores are due to differences in innate ability. It merely means that whatever causes children to differ in school performance also causes them to differ in test performance. Whether this difference reflects hereditary or environmental influences is an entirely separate question.

Also note that even defenders of IQ tests do not claim that the tests are perfect. At best, they measure one kind of intelligence, the kind of intelligence that helps people to do well in school. They do not measure the multiple kinds of intelligence that Gardner discusses and certainly not all the kinds of skills that any human culture might value.

CONCEPT CHECKS

8. A test of driving skills includes items requiring people to describe what they see. People with visual impairments score lower than people with good vision. Is the test therefore biased against people with visual impairments?
9. Suppose someone devises a new IQ test, and we discover that tall people generally get higher scores on this test than short people do. How could we determine whether this test is biased against short people? (Check your answers on page 341.)

How Do Heredity and Environment Affect IQ Scores?

The British scholar Francis Galton (1869/1978) was the first to offer evidence that the tendency toward high intelligence is hereditary. As evidence, he simply pointed out that eminent and distinguished men—politicians, judges, and the like—generally had several distinguished relatives. We no longer consider that evidence convincing, because distinguished people share environment as well as genes with their relatives. Besides, becoming a distinguished person is only partly a result of intelligence.

The question of how heredity affects intelligence has persisted to this day and continues to be difficult to answer. The issue is complicated because intellectual development undoubtedly depends on many genes (Chorney et al., 1998) and many environmental influences, each of which makes a small contribution. Here are descriptions of the available kinds of evidence, with their strengths and limitations:

Family Resemblances

Figure 9.13, based on a review of the available literature by Thomas Bouchard and Matthew McGue (1981), shows the correlations of IQ scores for people with various degrees of genetic relationship. These data are based almost entirely on European-American families; we do not know how well the results apply to other ethnic groups.

Because no IQ test has perfect reliability, a single individual taking the test on two occasions will get slightly different scores, and the correlation between the two test scores will be close to .9. The scores of monozygotic (identical) twins correlate with each other almost that well (Plomin & DeFries, 1980). Monozygotic twins resemble each other not only in overall IQ, but also in individual tests of verbal ability, spatial ability, processing speed, and memory; they continue to resemble each other throughout life—even beyond age 80 in one study (McClearn et al., 1997). Fraternal twins and nontwin siblings differ more than monozygotic twins do, but their scores are still fairly close. More remote relatives, such as cousins, have IQ scores that correlate positively but not strongly. A resemblance among relatives is consistent with the possibility of a genetic influence, but it is also consistent with the possibility of an environmental influence. (Closer relatives have more similar environment as well as more similar heredity.)

Note also in Figure 9.13 the positive correlation between unrelated children adopted into the same family. That correlation indicates a significant contribution from the environment. However, because this correlation is lower than the correlation between siblings (brothers or sisters) suggests that family environment does not account for all variations in IQ. (Alternatives include genetics, prenatal environment, and "nonshared" factors in the environment that differ from one family member to another.)

The comparison between monozygotic (identical) and dizygotic (fraternal) twins also shows a greater correlation between monozygotic twins than between dizygotic twins. This presumably reflects a genetic contribution, unless we can attribute these results to a tendency for monozygotic twins to spend more time together and to be treated more alike. However, researchers have found that identical twins who always *thought* they were fraternal twins resemble each other as much as other identical twins do; fraternal twins who always *thought* they were identical twins resemble each other no more than other fraternal twins do (Scarr, 1968; Scarr & Carter-Saltzman, 1979). That is, the main determinant of similarity in IQ is whether twins are actually identical, not whether they think they are identical.

Identical Twins Reared Apart

In Figure 9.13, note the high correlation between monozygotic twins reared apart. That is, identical twins who have been adopted by different parents and reared in separate environments nevertheless strongly resemble each other in IQ scores (Bouchard & McGue, 1981; Farber, 1981). That resemblance implies a genetic contribution to IQ. Skeptics point out that the "separate" environments have often been very similar (Farber, 1981; Kamin, 1974). In some cases, the biological parents raised one twin, and close relatives or

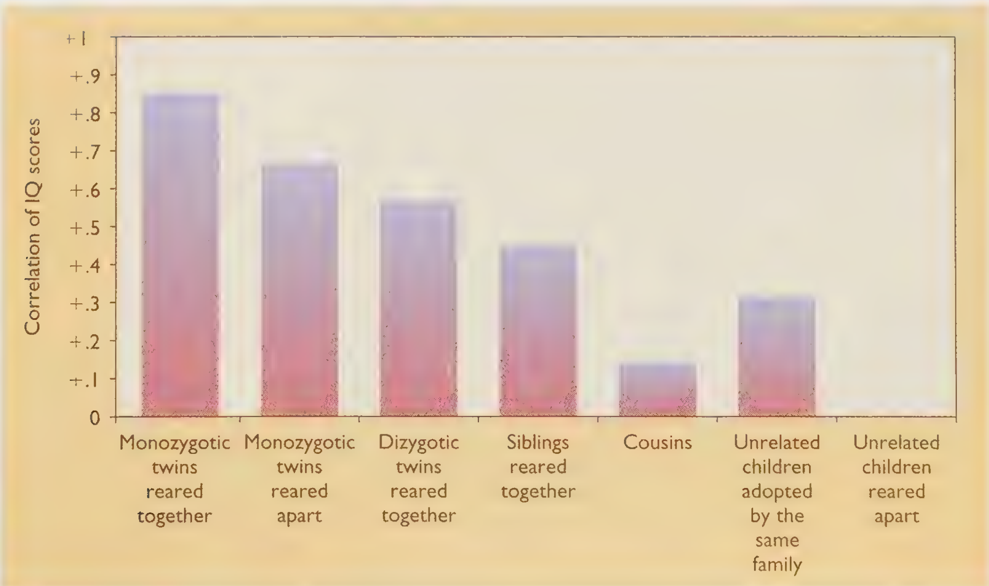


FIGURE 9.13 Mean correlations for the IQs of children with various degrees of genetic and environmental similarity. (Siblings are nontwin children in the same family.) (Adapted from Bouchard & McGue, 1981.)